

Surface Potential Decay on LDPE and LDPE/Al₂O₃ Nano-Composites: Measurements and Modeling

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ABSTRACT

Surface potential decay on LDPE and LDPE/Al₂O₃ nano-composite films charged by positive corona was studied. Potential distributions and decay on charged samples were measured using non-contact technique under ambient conditions. Results show that addition of 3 wt% of Al₂O₃ nano-filler drastically slows down the decay process, confirming this way modification of the charge transport process. The obtained decay characteristics are analyzed by considering various decay models, based on intrinsic conduction, dielectric polarization, surface charge de-trapping as well as charge injection and its transport. Applicability of the first three models is analyzed in terms of derived quantities such as material conductivity, potential decay rate and trap density distribution whereas the latter one is examined by means of computer simulations. Among the studied models, the ones based on dielectric polarization and charge transport provide best fits to potential decay characteristics.

Index Terms - Polyethylene, nano-dielectrics, surface potential, charge transport, dielectric polarization, trap density.

1 INTRODUCTION

EXPERIMENTAL and theoretical studies of surface charging and surface charge decay on polymeric materials have extensively been reported during the last decade [1-26]. The research has been stimulated by a necessity to understand fundamental processes involved in the electric charge transport phenomena and also by needs arising from practical applications of polymeric materials in such areas as electrets, electro-optics, surface modification, electrical insulation, nuclear physics and space industry [1-3]. In case of electrical insulation, main driving force for this type of studies was the development of new solid insulating materials for high voltage applications [17, 18, 21, 24, 25]. In most of these studies, surface of a flat sample is initially charged by external corona and thereafter the deposited charge is allowed to decay in an open circuit configuration (i.e., charged surface is open to air whereas opposite surface is in contact with a grounded metallic electrode) under the electric field created by the charges itself. The decay process is typically recorded by

measuring the change of surface potential. It is worth mentioning that most of the reported experiments were carried out on polyethylene specimens, especially low density polyethylene (LDPE), and on thin films with thickness in μm range [4-19]. A number of investigations on thick (mm range) samples were also reported [20-26].

Information on charge distributions on the surface and dynamics of its decay on thin films has been primarily utilized for determining material properties related to charge storage and its transport [5, 7, 9, 10, 17-19]. In case of thick samples, decay characteristics have been utilized either to evaluate charge retention capabilities of outdoor insulation materials and/or for estimating material properties such as volume conductivity [23-26]. Recently, surface potential decay measurements on nano-dielectric films were also reported [17, 18]. The results of these studies contribute mostly to understanding the impact of nano-particle addition on charge storage and transport in nano-composites.

In this paper we report on measurements of surface potential decay (SPD) on LDPE and LDPE/Al₂O₃ nano-composite samples of two different thicknesses. The measured potential

decay characteristics are analyzed and used to investigate applicability of various decay models presented in the literature. Some models are examined in terms of derived macroscopic parameters such as field dependent material conductivity and trap energy distribution while models based on charge injection and transport are analyzed by means of computer simulations.

2 SURFACE POTENTIAL DECAY MODELS

Surface charges deposited on a polymeric sample maintained in open circuit configuration can decay due to three main mechanisms, namely, bulk neutralization, surface conduction and gas neutralization (also referred to as surface neutralization [1]). It is worth mentioning that in this classification, dielectric relaxation, which also contribute to SPD [1, 3], is included in the bulk neutralization. Detailed discussions on these mechanisms can be found elsewhere [3, 20-22, 25].

Contributions of different mechanisms to the depletion of surface charges depend on particular conditions. Thus, gas neutralization can be minimized (even avoided) in SPD experiments by adjusting test procedure. As revealed in [3, 25], continuous positioning of Kelvin type electrostatic probe, which utilizes field nullifying technique for measuring surface potential, over charged area prevents arrival of gas ions to gas-solid interface. Charge leakage through the surface (surface conduction) may become significant if material is characterized by a high surface conductivity, for example due to aging or surface treatment [26] and/or highly uneven charge distributions giving rise to strong tangential field components. In addition, charge decay intensity is highly dependent on external conditions such as moisture content and presence of pollutants [23, 27]. As discussed in our earlier studies [25, 26] performed with non-uniform charge distributions, the influence of surface conduction can be easily estimated by observing lateral spreading of charged area during the SPD measurements.

If gas and surface neutralization is avoided, charge decay due to bulk leakage dominates. It can be due to one or more of three main mechanisms, namely, intrinsic conduction, charge injection and polarization [3, 20]. It is well established that when considering surface charge decay under open circuit configuration, resulting surface potential is governed by current continuity condition [1, 3]

$$\partial D / \partial t + \sigma E + \sum_i \mu_i \rho_i E = 0 \quad (1)$$

Here, the first term indicates time variation of the dielectric displacement D , which is dependent on polarization process inside the material since $D = \epsilon_0 E + P$, where E is the electric field strength, P is polarization, ϵ_0 is the permittivity of vacuum. The second term represents the current density due to intrinsic conductivity σ of the material. The third term describes the current density due to drift of injected charge carriers in material bulk. Here, μ and ρ are the mobility and the density, respectively, of particular type i of charge carriers. It is worth to recall that this formulation is only valid as long as the charge decay due to gas neutralization and surface conduction is prevented. In order to formulate a complete potential decay problem, it is essential to consider time dependent charge density together with the charge

injection from surface (if any) as a boundary condition on the gas-solid interface.

As almost all the earlier reported studies utilized surface potential measurements to characterize surface charge decay, it is important to establish a relation between these quantities. Assuming negligible space charges in the material bulk and uniformity of surface charging, a linear dependence between the potential V and the charge density q_s is obtained [25, 28]

$$V = q_s d / \epsilon \quad (2)$$

where d and ϵ stand for thickness and permittivity of the material, respectively. In case of strong non-uniformity of surface charge distributions and/or complex geometries of gas-solid interface, obtaining such a relation requires special calculation techniques [29].

2.1 MODEL BASED ON INTRINSIC CONDUCTION

According to (1), once polarization is stabilized and space charge effect can be neglected, the potential decay is mainly due to bulk conduction [1, 3] provided that deposited charges are deeply trapped in surface traps and the characteristic time of detrapping is long. Note that surface detrapping process may be essential for thin films as emphasized in [12]. For uniform charge distribution on surface of a flat material sample with large dimensions as compared to its thickness, the problem can be treated as a one-dimensional and intrinsic conductivity σ can be derived from SPD measurement as [24, 25]

$$\sigma = - \frac{\epsilon}{V(t)} \frac{dV(t)}{dt} \quad (3)$$

This indicates that a constant intrinsic conductivity should yield an exponential potential decay characteristic [1, 3]. However according to the definition, conductivity depends on density and mobility of charge carriers as $\sigma = \sum q \mu n$, where q is the elementary charge and summation is to be done for all types of charge carriers present in the material. Such definition presumes that both the densities of charged species n and their mobilities μ are not necessarily constant and may depend upon different factors, in particular, field magnitude, presence of trapping centers, etc. Hence, an idealized exponential SPD is hardly reachable in practice. Nevertheless, equation (3) may be employed to interpret potential decay shapes other than exponential one if certain hypotheses on charge generation and recombination in material bulk as well as on charge carriers' mobilities are introduced. For instance, conductivity in equation (3) can be assumed as field dependent, which allows interpreting SPD characteristics in a wide time interval when the field strength in the material is reduced significantly. Such approach has been employed in [30] where the potential decay was assumed to be associated with continuous charge neutralization at the surface due to bulk conduction. The field dependence of the conductivity is taken into consideration by applying a modified Poole-Frenkel model. The approach was utilized to reproduce the potential decay characteristics presented in [9] and it was concluded that the intrinsic conductivity model was sufficient to verify experimentally observed potential decay dynamics. Recently,

the same procedure was applied in [24, 25] to study measured SPD on thick polymeric samples. In all these cases, field dependent conductivity $\sigma(E)$ was represented as exponential function of the square root of the electric field (or surface potential):

$$\sigma(E) = \sigma_0 \exp(\beta \sqrt{E}) \quad (4)$$

$$\beta = \frac{q}{kT} \sqrt{\frac{q}{\pi\epsilon}} \quad (5)$$

Here, β is Poole-Frenkel coefficient, k is Boltzmann's constant and T is the absolute temperature.

In [24], calculated and measured β coefficients were compared for few types of materials and a fair agreement was found in case of those containing small amount of additives. In contrary, according to [25], even though the conductivity of some silicone rubber types showed an exponential variation with $\sqrt{E}$, their conductivity did not obeyed Poole-Frenkel dependence, at least as regards to the theoretical value of β coefficient.

In [31-33], an approach similar to the intrinsic conductivity model was used to calculate the dynamics of surface charge density. In these studies, each charge decay mechanism, including surface conduction and gas neutralization, was represented by corresponding equivalent conductivity or current source. However, the applicability of such a model requires further analyses.

2.2 MODEL BASED ON DIELECTRIC POLARIZATION

Presence of electric charges on a dielectric surface results in modifications of electric field inside the material and activates polarization processes, yielding in formation of additional interfacial charge layers. Interaction of these charges with the pre-deposited charge is still not completely understood. It may however affect the surface potential decay. Under the assumptions of zero conductivity ($\sigma = 0$) and no space charge ($\rho = 0$) (i.e. no charge injection), equation (1) yields a constant surface charge density. In such situation, the potential decay can be described solely by a time dependent dipolar polarization [1, 3]. A characteristic feature of almost all insulating polymers is that dielectric relaxation is very slow, being a consequence of internal molecular reorganization and complex interfacial phenomena taking place due to the existing electric field distribution [1]. Despite of many reported studies where polarization was neglected (either with or without valid reasons), some authors (see e.g. [1, 3]) highlighted the importance of taking this phenomenon into account. According to the recent review [1], polarization leads to power law relation between potential decay rate (dV/dt) and time (t), which appears as linear segments or asymptotes in log-log plot of (dV/dt) vs (t). However, it is worth mentioning that similar relationships can be observed due to dispersive transport as well [1]. More detailed mathematical formulation and utilization of polarization model to analyze surface potential decay in polymers can be found in [3, 20, 27].

2.3 MODEL BASED ON CHARGE DETRAPPING

If carrier's transit time is low compared to their mean characteristic de-trapping time, the potential decay becomes

solely controlled by the de-trapping processes. Demarcation energy level, E_d represents the border between emptied states and filled states so that states located above E_d are emptied during the time spent after charge deposition, whereas states below E_d still remain frozen. With the time passing, E_d moved down (in energy terms) and more states become emptied [34]. As a result, the potential dynamics is governed by the progressive change of trap energy depth. If the decay is dominated by progressive detrapping and retrapping is negligible, a plot of tdV/dt vs. demarcation energy E_d provides an image of trap density distribution [1, 3, 24, 26].

In [35], the demarcation energy model was used to obtain such image for bulk states in thin polymeric films. The same approach was utilized in [3] to treat results of surface potential decay measurements on corona charged thin films, where charges were initially trapped in surface states. Application of the model has been extended in [24, 25] for attaining surface trap density distributions in thick polymeric samples charged by corona. In [36], the same model was utilized to obtain absolute trap densities in different polymeric materials based on isothermal surface potential decay measurements.

2.4 MODEL BASED ON CHARGE INJECTION AND TRANSPORT

Some of the proposed models of surface potential decay assume that the deposited charges are injected through the gas-solid interface (i.e. charged surface) into material bulk and are further transported inside it. This model was attractive for analyzing the decay process e.g. in thin films where field strength is high enough for maintaining charge injection. One of the reasons of using this approach in numerous studies is the possibility to explain part of the observed decay characteristics including cross-over phenomenon [4, 9]. The mathematical representation of these models is achieved by utilizing continuity equations for charge carriers, coupled with Poisson's equation for the field. In [37], a model based on volume conduction due to drift of charges was developed, taking into account both time and spatial variations of space charge. In [38], similar model was extended to account for charge diffusion. Another version of injection and transport models was developed by considering instantaneous partial injection and field independent mobility of injected charges and it was used to study transient behavior of charge carriers generated by pulse illumination of a flat insulator sample [39]. Later, the same model was further modified to include trapping mechanisms in the bulk [40]. In [41], a more sophisticated approach was presented, which accounted for field dependent mobility and either instantaneous or time dependent injection. The model was used to simulate experimental potential decay data presented for polyethylene in [4]. The performed comparison showed that the model results reflected experimentally observed general features, such as initial steep decay at high initial surface potentials, almost flat decay curves for lower initial potentials and cross-over phenomenon. Further theoretical studies suggested that all surface charges are injected into the bulk within very short time at high initial potentials whereas for low initial potentials, little or no charge injection took place. For

intermediate potentials magnitudes, injection process was considered to be time and field dependent. Further progress in developing computer models of surface charge relaxation has been achieved by introducing trapping and de-trapping processes to the transport model [9]. In [11], an approach similar to [9] was reported and modified by introducing a simple power law relationship to describe field dependent mobility of charge carriers.

A bipolar charge injection and transport model in which Schottky type injection was implemented both from metal-solid and gas-solid interfaces was presented in [42]. This type of model (without considering open gas-solid interface) has been extensively used to study charge transport and space charge buildup in insulating polymers, especially in polyethylene [43-45]. In [15], the necessity of considering bipolar charge injection for modelling SPD was highlighted owing the experimental evidences of presence of homo-charges close to both interfaces. Thus, Schottky emission was assumed while introducing field dependent mobility in form of a power law relation. Even though the validity of such representation of mobility is not convincing, the model showed a capability of simulating the cross-over of potential decay curves on 50 μm thick LDPE films. Another work of the same authors [14] considered Fowler-Nordheim tunneling instead of Schottky mechanism of emission to describe charge injection at gas-solid interface.

In a very recent work [19] a bipolar charge transport model is utilized to study potential decay on negatively charged LDPE films. It assumes that all the ions reaching the surface during the charging transfer their electrons to surface traps and those trapped electrons are released into the bulk during the decay process. The model has been supported by a comparison of the results with analytical calculations performed based on unipolar charge transport presented in [41]. Thus, a generalization of the ideas presented in [19] still requires verifications.

The above presented considerations indicate that despite of the fact that charge injection and transport over gas-solid interface have been studied over several decades, there still remain uncertainties related to the mechanism of charge transfer. In addition, the uncertainties also concerning the validity of various model parameters, for example, the field dependence of carriers' mobility.

3 EXPERIMENTS

3.1 MATERIALS SAMPLES

Three types of material specimens were used in this study as presented in Table 1, namely, pure LDPE, LDPE with 0.02 wt% antioxidant (AO) and LDPE with 0.02 wt % AO and 3 wt % Al_2O_3 nano-filler, which were manufactured in the Polymer Technology laboratory of the Royal Institute of Technology in Stockholm, Sweden [46]. The samples used had an average surface area of about 70 mm $\times$ 70 mm and thicknesses of 80 and 150 μm . The manufacturing procedure of Al_2O_3 filled samples included grinding LDPE pellets into a powder (~ 0.5 mm in size) after being immersed in liquid

nitrogen for 1 h. After that, a certain amount of Al_2O_3 nanoparticles (0.3 g) was dispersed in heptane (3 ml) using a polypropylene tube. Next, Irganox 1076 (2 mg) dispersed in heptane (0.3 ml) was added into the suspension which was further treated in an ultrasonic bath for 5 min. Then, polyethylene powder (9.7 g) was added and the suspension was shaken for 1 h. After that, the powder was dried in an oven at 80 $^\circ\text{C}$ overnight to evaporate the solvent. At last, the dry mixture powder was compounded by thermal extrusion at 135 $^\circ\text{C}$ in 6 min. The rotation speed of the extruder was 250 rpm.

3.2 EXPERIMENTAL SETUP AND PROCEDURE

A schematic diagram of the experiment setup is shown in Figure 1. The sample was placed on a grounded copper plate mounted on a rotating table, while its upper surface was kept open to the surrounding air. Corona charging of the sample was achieved by using a needle (hemispherical tip of diameter 0.95 mm) and a grid electrode (9 cm $\times$ 9 cm) mounted on a fixed arm. Surface-to-grid and needle-to-grid distances were 5 mm and 2 mm, respectively. The sample surface was first charged for 5 min by applying high potential to the needle and the grid. Two combinations for needle/grid potentials were utilized, namely, +5.6 kV/+2.1 kV and +6.6 kV/+3.1 kV. Upon completion of the charging, the table was rotated to bring the center of the sample directly under a Kelvin type electrostatic probe (Trek PN 6000B) connected to an electrostatic voltmeter (Trek 347). The time lag between completing the charging and starting surface potential measurements was less than 5 s. A computer equipped with a data acquisition card was used to record the measured data. During the decay measurements, the energized probe was always kept above the sample center to avoid effect of gas neutralization. At several cases, potentials at points on the surface other than the center were also recorded to map its distribution on the surface at selected time instants. All the experiments were conducted at ambient conditions (temperature 19-21 $^\circ\text{C}$, air pressure 995-1005 hPa and relative humidity 31-35%).

Table 1. Details of studied material types.

Material type	Composition	Thickness / (μm)
1	LDPE	150
2	LDPE+ antioxidant (AO)	150
3		80
4	LDPE+AO+ Al_2O_3 nano-filler	150
5		80

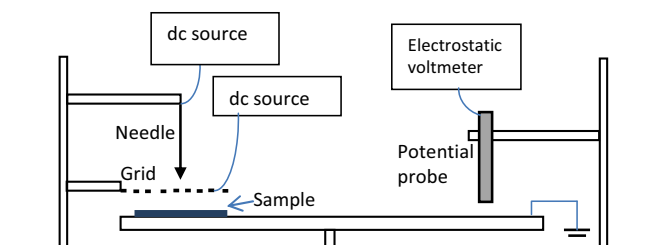


Figure 1. Sketch of experimental set-up.

3.3 SURFACE POTENTIAL DISTRIBUTIONS

Typical potential distributions measured during decay are shown in Figure 2 for several selected time instants. From the potential profile recorded 5 s after the charging (referred as initial potential below), it is evident that the use of the grid electrode allows for efficient control of ionic flux produced by corona from the needle and the resulted surface potential distribution induced by deposited charges is almost uniform. The magnitude of the initial potential is close to the potential applied to the grid. During the decay, the distribution remains uniform without any lateral expansion of the charged area that reflects negligible contribution of surface conduction.

3.4 SURFACE POTENTIAL DECAY

Surface potential decay on samples charged at +3.1 kV grid voltages is shown in Figure 3. Note that all samples were initially charged to the same potential. As seen, the decay characteristics for pure LDPE and LDPE+AO of thickness of 150 μm are practically identical indicating no influence of the small amount of antioxidant. The faster decay on LDPE+AO sample of thickness of 80 μm can be related to higher electric field strength induced by deposited charges in material bulk. It is notable that the addition of Al_2O_3 nano-particles to LDPE slows down the decay significantly irrespective of sample

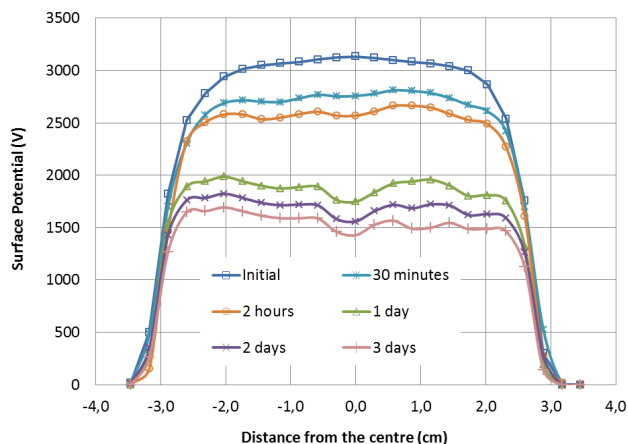


Figure 2. Surface potential distributions on Pure LDPE sample surface.

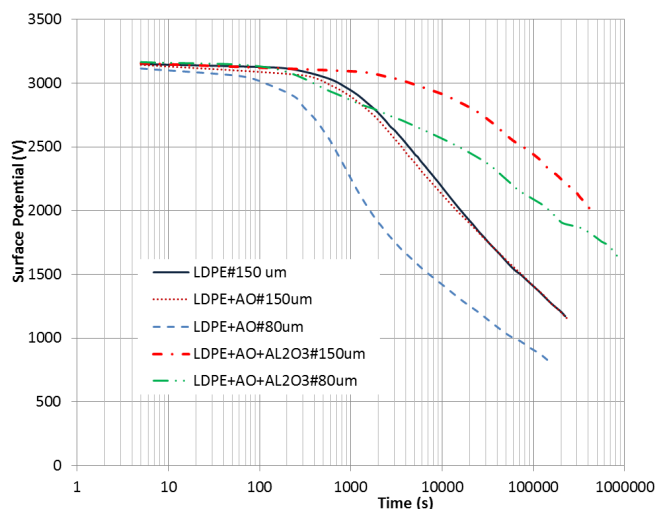


Figure 3. Surface potential decay on LDPE samples charged by applying +3.1 kV to the grid and +6.6 kV to the needle.

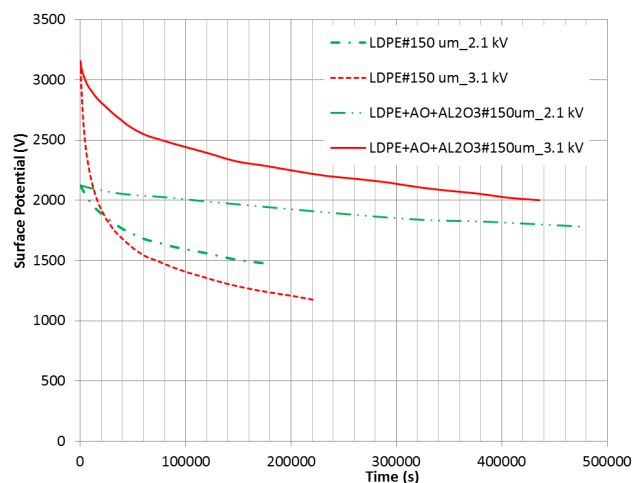


Figure 4. Potential decay characteristics at different initial potentials; results are shown for 150 μm thick samples.

thickness. Thus for 150 μm nano-filled sample, the surface potential drops from 3.1 kV to 2 kV after 5 days whereas such decay took only 3 hours in case of the pure material.

The influence of charging conditions on decay process can be observed in Figure 4, where the decay characteristics obtained for initial potentials +2.1 kV and +3.1 kV are compared. Note that results obtained with the LDPE+AO sample are not shown for clarity as similar decay as for pure LDPE was recorded. In the plot, one can clearly observe a cross-over of the decay curves for the pure material. Such behavior for pure LDPE may be attributed to nonlinear space charge effect. In case of nano-filled material, such cross-over can be expected (following the trends of the curves), however, was not reached in the experiments even after considerably long decay measurements (almost 6 days). Similar characteristics were observed for 80 μm thick samples.

4 APPLICATIONS OF SPD MODELS

As mentioned in the previous chapter, contributions of both surface conduction and gas neutralization to surface charge decay are negligible in the present study. Thus, four different types of models reviewed in section 2 can be employed for analyzing the obtained data. Three models based on macroscopic approach, namely, intrinsic conduction, de-trapping and polarization models are presented in this section with regards to the decay measurements at initial potential +3.1 kV. Application of the charge injection and transport model is presented in the next chapter.

4.1 MODEL BASED ON INTRINSIC CONDUCTION

The cross-over of the characteristics measured for LDPE (Figure 4) indicates that different decay rates dV/dt correspond to the same potential thus providing different values for the conductivity according to equation (3). Hence, the intrinsic conduction model is not directly applicable to LDPE used in the present study under considered charging and decay conditions. It, however, should be considered for the nano-composite. An analysis of materials conductivity with regard to the widely used charge transport models, i.e. Poole-Frenkel and hopping [47], is provided below.

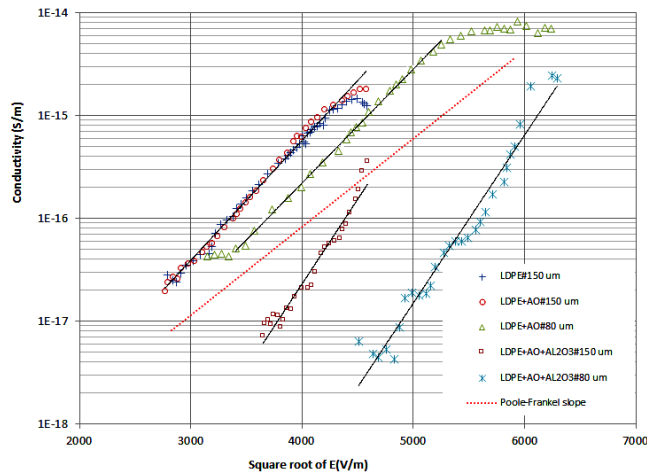


Figure 5. Equivalent conductivities calculated from measured SPD characteristics (initial potential 3.1 kV). Solid lines represent exponential fits of the experimental data. The dotted line indicates the slope corresponding to β coefficient (5) obtained using relative permittivity 2.3.

4.1.1 POOLE-FRENKEL CONDUCTION

Materials equivalent conductivities calculated according to equation (4) using the measured surface potential decay rates are plotted in Figure 5 as function of $\sqrt{E}$. The dotted line corresponds to theoretical values obtained with theoretical Poole-Frenkel coefficient given in equation (5). As seen, variations of the conductivity of these materials show a Poole-Frenkel type of behavior (except for high field region in case of 80 μm thick LDPE+AO). However, the slopes of the experimental dependences deviate from the theoretical one. A better match can be obtained by assuming the value of relative permittivity in the range 0.9-1.4, which however is not realistic. Thus Poole-Frenkel conduction model does not seem to be applicable for describing the observed decay process.

4.1.2 HOPPING CONDUCTION

Hopping conduction refers to a transport of charge carriers, which are released from trapping sites by thermal activation and move to available empty states [47]. The corresponding equivalent conductivity is expressed as [47,48]

$$\sigma(E) = \frac{2q\nu N a}{E} \exp\left(-\frac{q\phi}{kT}\right) \sinh\left(\frac{qaE}{2kT}\right) \quad (6)$$

where a represents average hopping distance, N is carrier density, ν is attempt to escape frequency, and ϕ is average hopping barrier height. The results of the fitting of the conductivities deduced from SPD measurements by hopping conduction model are shown in Figure 6 by solid lines. Best fitting was found when assuming the average barrier height $\phi = 0.65$ eV, which is in the range reported for LDPE [49]. The values of a and N used for the fitting are provided in Table 2 (note that the average hopping distance is close to 15 nm, as suggested in [49] for LDPE). As seen from Figure 6, the theoretical dependences describe experimental results reasonably well except for, again, those obtained with 80 μm thick LDPE+AO sample in the high field region (above ~ 27 kV/mm). Almost constant conductivity observed for the latter case suggests an ohmic type of conduction with an intrinsic

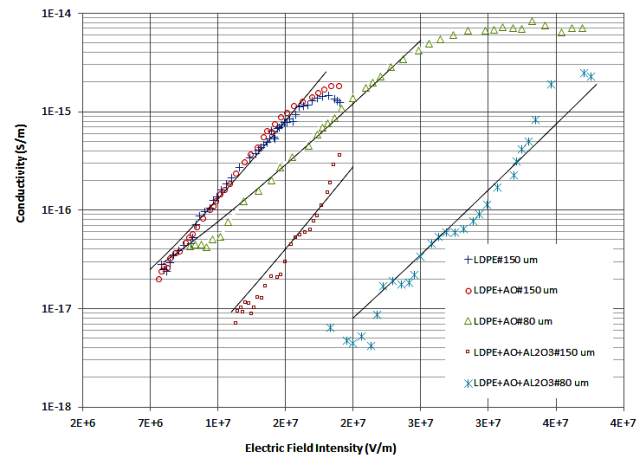


Figure 6. Equivalent conductivities calculated from measured SPD characteristics (initial potential 3.1 kV) fitted by hopping model given in equation (6).

Table 2. Parameters of hopping conduction model.

Material type	Average hopping distance a (nm)	Carrier density N (m^{-3})
LDPE	22	6×10^{13}
LDPE+ AO#150 μm	22	6×10^{13}
LDPE+ AO#80 μm	17	1×10^{14}
LDPE+ AO+AL ₂ O ₃ #150 μm	22	3×10^{12}
LDPE+ AO+AL ₂ O ₃ #80 μm	17	1×10^{12}

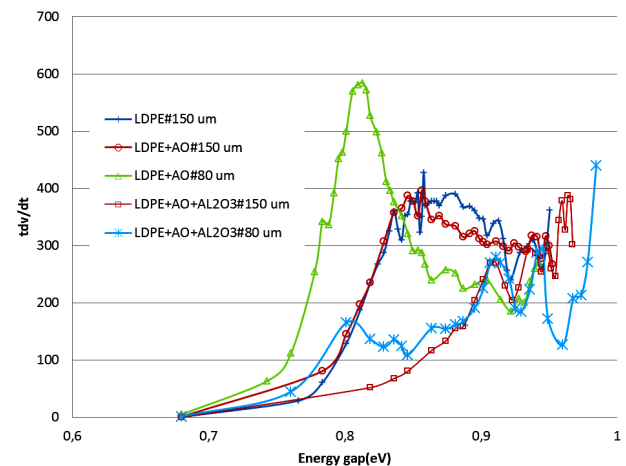


Figure 7. Image of trap energy distribution deduced from SPD.

conductivity of $\sim 7 \cdot 10^{-15}$ S/m. Thus, the hopping conduction model may be utilized to elucidate the observed slow potential decay on the nano-filled material (Figure 3) which may be attributed to a reduction of the number of charge carriers released from traps.

4.2 MODEL BASED ON CHARGE DETRAPPING

As mentioned, the SPD model based on charge detrapping allows for obtaining an image of trap energy distribution [1, 3, 24]. Such distributions deduced from the measured decay characteristics are shown in Figure 7. In the calculations, the attempt to escape frequency $\nu = 10^{11} \text{ s}^{-1}$ was assumed [24, 34].

As seen, the spread of trap energy is between 0.7 and 1 eV for the studied materials, which is quite common for LDPE [44]. A peak density appears at energy level 0.8-0.9 eV for LDPE without nano-particles. Apart from that, the distribution obtained for 80 μm thick sample is narrow as compared to the profiles attained for 150 μm thick samples. Even though exact reason is not clear, it may be due to different conditions during manufacturing of the films. The addition of nano-filler leads to significant widening of the distributions and shifting them to higher energies. The distinct peaks observed for pure material are not visible for nano-composites, at least for the considered SPD period. Thus, introduction of nano-particles increases the range of energies of traps that leads to enhanced charge accumulation in the material. This may also explain the reduced magnitudes of the conductivities of the nano-composites (Figures 5 and 6).

4.3 MODEL BASED ON DIELECTRIC RELAXATION

As discussed in [1], potential decay under the influence of slow dielectric polarization may result in SPD rate curves being composed of time power law(s) which appear as linear segments or asymptotes on $\log(dV/dt)$ vs $\log(t)$ plot. Such dependences obtained for the studied materials are shown in Figure 8. It can be clearly observed that for each material type they can be fitted by two linear segments with knee points close to 1000 s. The segments at longer time instants demonstrate slopes of -1 for LDPE samples, irrespectively of the thickness. When the material is filled with nano-particles, the knee point appears earlier and the slope changes from -1 to -(0.7-0.9). When considering these slopes, one should note that the observed knee point is not the transit point, which is usually referred to in the concept of the dielectric polarization model [1].

The “ $\log(dV/dt)$ vs $\log(t)$ ” characteristics indicate a contribution of a slow polarization process to SPD mainly at later stage of the decay. When considering this behavior for pure LDPE, one should recall that it is a non-polar material and typically shows low dielectric losses without any significant relaxation mechanisms [50]. For LDPE-based nano-composites not much information is available yet, but evidences for appearance of relaxation peaks at both high and low frequencies have recently been reported [50]. These specific features should be considered as the other decay mechanisms, such as surface de-trapping and charge injection and transport, and may also result in similar power law relations between the decay rate and time [3]. One way to distinguish the impact of polarization from the other mechanisms can be to relate the transit time to information provided by dielectric response measurements that require additional extended studies.

5 CHARGE INJECTION AND TRANSPORT MODEL OF SPD

As already mentioned in chapter 2, processes leading to injection of charges on gas-solid interfaces are not well understood and quantified. Moreover, not only charge injection mechanism but even types of charge carriers appearing in the solid due to deposition of ions generated in

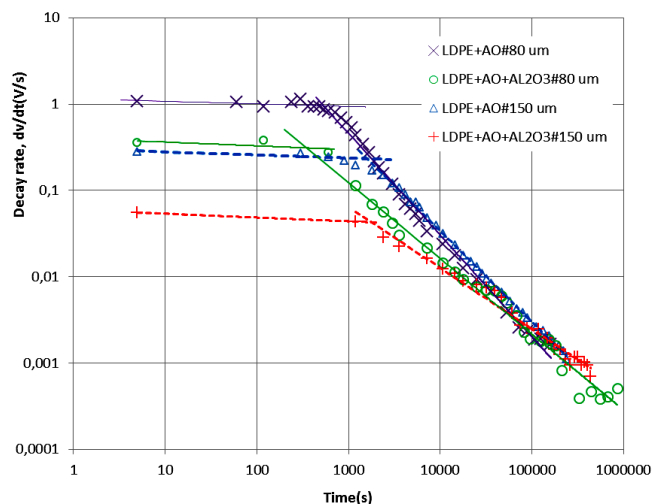


Figure 8. Variation of the decay rate with time during potential decay on samples charged to 3.1 kV.

air by corona are not well explained. In case of metal-dielectric interface, Schottky model is widely utilized even though it often requires adjusting model parameters to provide reasonably good quantification of the injection process [47]. At field strength higher than $\sim 10^8$ V/m, Fowler-Nordheim mechanism is usually considered. However, adopting these models to charge injection on gas-solid dielectric interface is not feasible due essentially different physical process involved, which should account for charge transfer from/to deposited air ions and intrinsic carriers in the solid material. Therefore in the present study, rather than using the injection models, it is assumed that ionic species approaching material surface from gas site are trapped in surface traps and no charge injection into solid bulk occurs at gas-solid interface. Since positive corona is used, positive ions are accumulated on the surface and form a charged layer inducing electric field in the bulk that stimulates injection and transport of electrons from metallic surface on the opposite side of the sample (recall that it lays on a grounded copper electrode). While reaching the gas-solid interface, the electrons recombine with the ionic species, reducing this way the amount of the deposited charge. Consequently, the model of charge transport is reduced to a unipolar case and accounts for drift and diffusion of charge carriers injected from the metallic electrode only.

5.1 MATHEMATICAL DESCRIPTION

In this study, so-called drift-diffusion approximation of transport of electrical charges in the materials is utilized (details can be found elsewhere [42-45]). Time and space variations of the density of electrons in the conduction band n_e are described by continuity equation (7) for their total flux that reflects mass conservation principle. It is assumed that the drift and diffusion of the charge carriers through the material bulk are associated with their trapping and de-trapping. The dynamics of the trapped charge density n_{etr} is described by an additional differential equation (8) accounting for corresponding rates of the process. Poisson's equation for the

electric potential (i.e. equation (9)) is used for obtaining local field distribution in the material. All the three equations are coupled through field dependences of kinetic and rate coefficients, surface and net space charge densities:

$$\partial n_e / \partial t + \nabla \cdot (-n_e \mu_e \mathbf{E} - D_e \nabla n_e) = DTR_e - TR_e \quad (7)$$

$$d n_{etr} / d t = -DTR_e + TR_e \quad (8)$$

$$\nabla(\epsilon_r \epsilon_0 \nabla \phi) = -q(-n_e - n_{etr}) \quad (9)$$

Here, μ is the mobility of electrons; D_e is the diffusion coefficient; $\mathbf{E}$ is the vector of the electric field; ϕ is the electric potential; and t stands for time. The trapping TR_e and de-trapping DTR_e rates are defined as

$$TR_e = B_e n_e (1 - n_{etr} / n_{etr0}) \quad (10)$$

$$DTR_e = v n_{etr} \exp(-\phi_{etr} / kT) (n_{etr} / n_{etr0}) \quad (11)$$

Here, B_e is trapping coefficient for electrons; ϕ_{etr} is energy barrier heights for de-trapping of electrons; n_{etr0} is the density of traps.

The decay of the deposited surface charge is obtained by solving equation (12) defined on gas- solid interface:

$$dq_s / dt = -J_{out} \quad (12)$$

where q_s stands for the surface charge density and J_{out} represents current density induced by charges leaving open surface of the sample.

Charge injection at the metal-solid interface is considered to be of Schottky type [42-45] and corresponding current density is introduced as

$$J_{SK} = AT^2 \exp\left(\frac{-q\phi_{inj}}{kT}\right) \exp\left(\frac{q}{kT} \sqrt{\frac{qE_s}{4\pi\epsilon_r\epsilon_0}}\right) \quad (13)$$

Table 3. Typical values of coefficient and parameters for LDPE [44].

Parameter	Value
Electron mobility * μ_e , $\text{m}^2\text{V}^{-1}\text{s}^{-1}$	1×10^{-14}
Diffusion coefficient D_e , m^2s^{-1}	$(kT / q) \mu_e$
Trapping coefficient B_e , s^{-1}	0.1
Density of traps * n_{etr0} , m^{-3}	6.25×10^{20}
Barrier height for de-trapping ϕ_{etr} , eV	0.96
Attempt to escape frequency ν , s^{-1}	6×10^{12}
Barrier height for injection of electrons from cathode ϕ_{inj} , eV	1.15-1.27 (1.17 was used)
Relative permittivity of solid material ϵ_r	2.3

* used in parametric study.

Table 4. Values used in the parametric study.

Test case			Density of deep traps n_{etr0} , m^{-3}	Electron mobility $(\text{m}^2\text{V}^{-1}\text{s}^{-1})$
Material	Thickness (μm)	Initial potential (kV)		
LDPE and LDPE+AO	150	3.1	6.25×10^{20}	6×10^{-13}
		2.1		0.2×10^{-13}
	80	3.1		3×10^{-13}
		2.1		1.2×10^{-13}
LDPE +AO +Al ₂ O ₃	150	3.1	6.25×10^{21}	1×10^{-13}
		2.1		0.3×10^{-13}
	80	3.1		0.7×10^{-13}
		2.1		0.4×10^{-13}

where $A = 1.2 \cdot 10^6 \text{ Am}^{-2}\text{K}^{-2}$ is Richardson's constant; ϕ_{inj} is the barrier height for injection, T stands for temperature, and E_s is the field strength at the surface.

As potential distributions on samples surfaces recorded in the experiments were uniform during the decay (Figure 2), the model was simplified to one dimensional case and equations (7) - (13) were implemented in COMSOL Multiphysics package. The model parameters were adopted following [44], as shown in Table 3. In the calculations, the mobility and density of traps were used as parameters to fit the measured potential decay curves and their values providing the best fit are shown in Table 4. It should be noticed that the obtained values are close to those reported in the literature for polyethylene.

5.2 RESULTS OF THE SIMULATIONS

A comparison of the simulated and measured potential decay characteristics for LDPE samples without nano-filler is presented in Figure 9. It can be seen that the proposed unipolar charge transport model is capable of predicting the decay characteristics with a good accuracy, except for 80 μm thick sample charged up to +3.1 kV. In this case, an

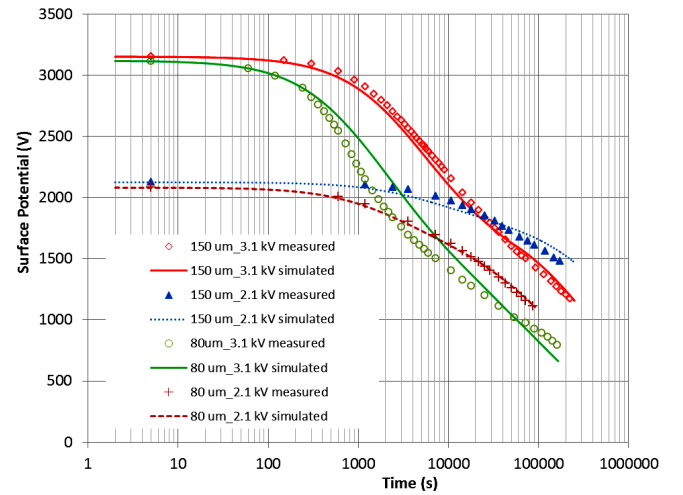


Figure 9. Measured and simulated SPD characteristics for LDPE.

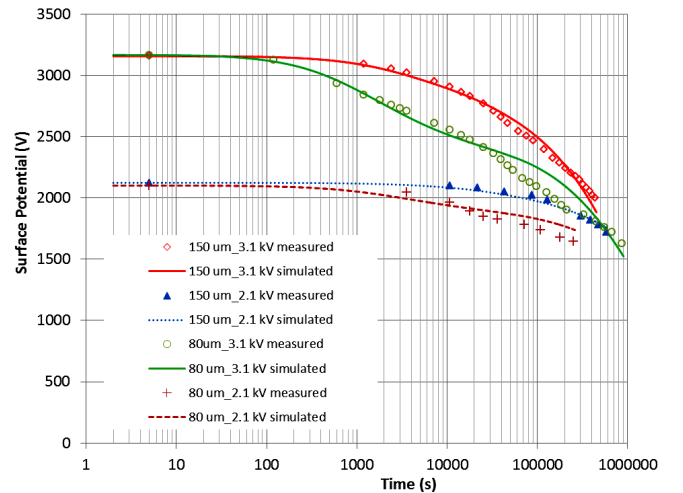


Figure 10. Measured and simulated SPD characteristics for nano-composite.

agreement with the experimental data was achieved at short and long times while a deviation (maximum ~10%) is observed at intermediate times. This can possibly be attributed to uncertainties in determining electron mobility, which may vary with sample thickness as well as the initial potential (Table 4).

The simulation conducted for the LDPE-based nano-composite yielded SPD characteristics that well agree with the experimental results, Figure 10. However, similarly to the case without nano-fillers, the computed SPD curve for 80 µm thick sample charged to 3.1 kV deviates from the measured one at the later stage of the decay. It is notable also that the shape of the decay characteristic in this case is considerably different from the others and required introducing an increased trap density level (by one order of magnitude, see Table 4) to fit the experimentally obtained dependence. Such an increase may be justified in the case of nano-composite, where enhanced charge trapping and reduced electron mobility are observed.

6 CONCLUSION

Surface potential decay characteristics were measured on films of LDPE with and without antioxidant as well as on films of LDPE/Al₂O₃ nano-composite. It was found that the addition of 0.02 wt% antioxidant did not influence decay process. At the same time, the addition of 3 wt% of Al₂O₃ nano-particles to LDPE drastically slowed down the charge decay process irrespectively of sample thickness. The obtained results were analyzed by comparing the experimental decay curves with different model simulations based on intrinsic conduction, dielectric polarization, surface de-trapping, and charge injection and transport.

The analyses showed that the model based on intrinsic conduction was not capable to predict the experimentally observed cross-over phenomenon in LDPE. However, material's conductivity deduced from this model pointed to a dominating role of hopping conduction in the decay process rather than Poole-Frenkel conduction. The model based on dielectric polarization indicated a possible contribution of this mechanism at the later stage of the decay. However, the applicability of this model still needs to be verified by means of extended dielectric response studies. The de-trapping based model yielded images of trap energy distributions and showed widening of the trap energy profile and its shift towards higher energy levels with the addition of nano-filler. The charge transport model based on the assumption of zero charge injection into the solid material from the charged gas-solid interface was capable of reproducing most of the features of the experimental decay curves. The agreement between computed and measured results was achieved by adjusting levels of electron mobility and trap density in the investigated materials.

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